

## Phase 1: System Planning and Distribution Selection (Week 1)

1. Create a System Architecture Diagram showing both systems and network connections?

Ans: An administrator workstation and a headless Ubuntu Server are each of the separate systems that make up the client-server architecture used for operating system deployment. An Ubuntu Server hosts platform utilities, or an admin server utilises it in remote support. A Virtual Box Host Only personal network allows the systems to link using one another while still free of other networks. The current Secure Shell (SSH), which protects the entire process, is used for running the systems online. The computer running Ubuntu has been given the private IP address 192.168.56.101 for the connection between users and the server.

2. Distribution Selection Justification comparing your chosen server distribution with alternatives?

Ans: Ubuntu Server 24.04 Long- Term Support (LTS) is an OS the fact was chosen for usage on the project because its support, capacity, and control were taken into factors. The Ubuntu Long Term Support (LTS) operating system has a long support term. On stability, one of the strengths of Ubuntu Server is that its foundation is on the Debian platform, which is dependent on newer and proven software. This is achieved through a controlled release by Canonical. This is remarkably beneficial to admins since they will be able to reap the benefits of newer technological developments without having an adverse effect on the stability of the actual server.

At the forefront of Debian is ensuring that its server is extremely stable. This is attained by ensuring that software is frozen. This has, at times, caused users to possess outdated software. Another option was also CentOS, which was also taken as an alternative. CentOS had generally always been widely used because of its ability to provide compatibility for Red Hat Enterprise Linux, commonly known as RHEL. However, due to the emerging changes in the release of CentOS, known as CentOS Stream, it has not yet reached its peak in terms of predictability.

There are also emerging alternatives that could potentially replace CentOS, which include Alma Linux and Rocky Linux, but compared to Ubuntu Server, community resources are not elaborate. Ubuntu Server is well-documented, and its community is loyal and very popular within the industry. Ubuntu Server is also well-supported by modern security software and very popular within academic and cloud computing settings because of its high compatibility with security software and virtualization software. As far as operating systems are concerned, the most appropriate one that can serve as an example in the critical areas involved in the workings of an OS, which are process scheduling, memory management, file system structure, networking, and access control mechanism, is Ubuntu Server OS. Ubuntu Server OS is the most

appropriate and genuine one among all other server versions available for selection in terms of functionality and compatibility with stability and the latest version.

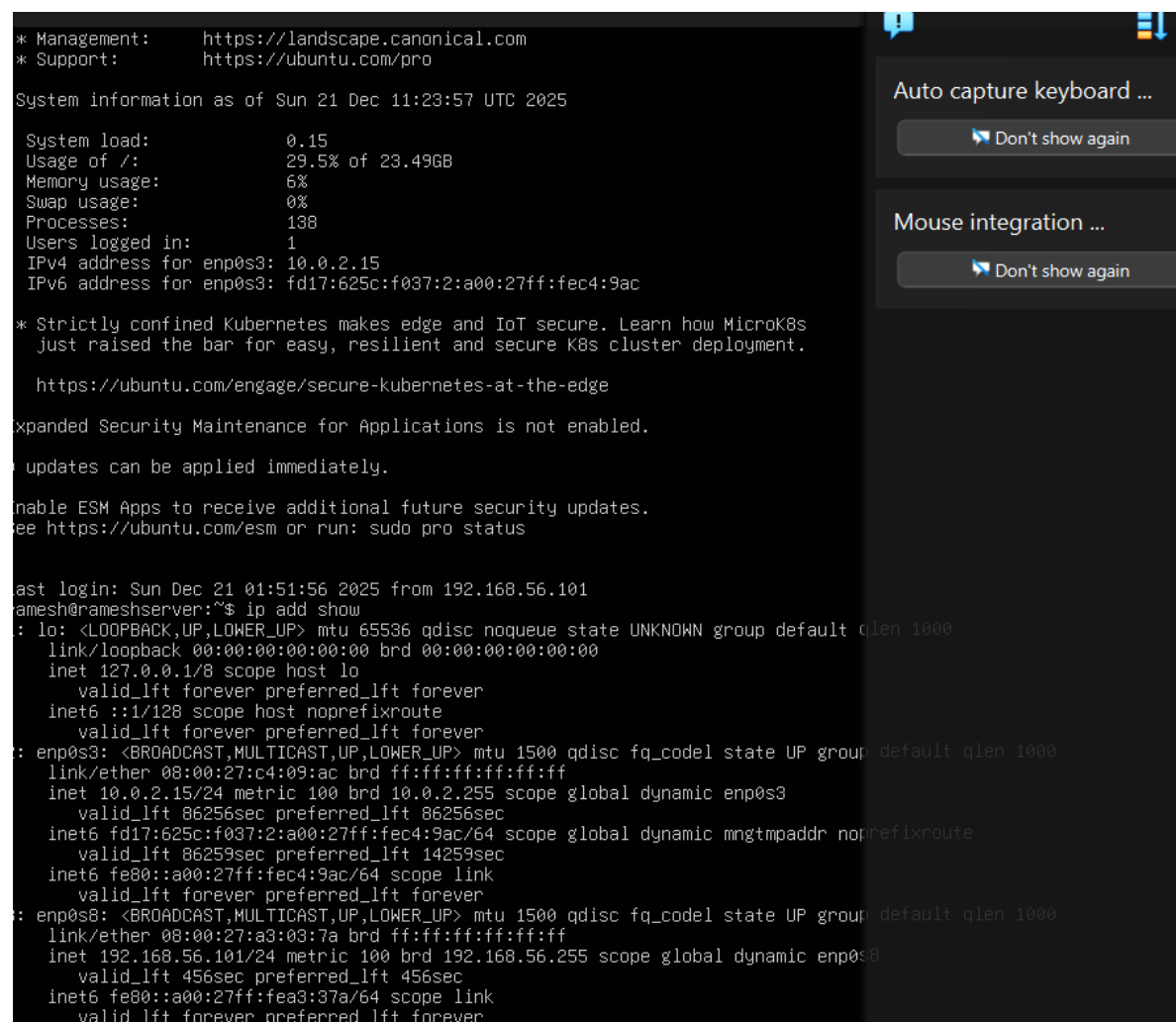
### 3. Workstation configuration decision justifying your choice of workstation option?

Ans: A different workstation was selected for computer management tasks to implement guidelines with control of the system. By removing the need to run a GUI from the server, controlling it via SSH decreases the server's power needs while leaving it less open to attacks.

The workstation provides a setting for safely entering commands, checking performance, and managing file systems. Each of those acts complies with the least rights ideal. Working on a workstation highlights commands ability and remote workstation operation, both of which are key capacities as will be learnt during this module.

### 4. Network configuration documentation covering VirtualBox settings and IP addressing?

Ans:

A screenshot of a terminal window with a dark background. The terminal displays system information and network configuration. On the right side of the terminal window, there are two panels: 'Auto capture keyboard ...' and 'Mouse integration ...', both with a 'Don't show again' button. The terminal output includes management and support links, system load and usage statistics, IP addresses for enp0s3, and a detailed network configuration for three interfaces: lo, enp0s3, and enp0s8. The configuration for each interface includes link type, MTU, queue discipline, state, group, and various IP addresses with their metrics and scopes.

```
* Management:      https://landscape.canonical.com
* Support:         https://ubuntu.com/pro

System information as of Sun 21 Dec 11:23:57 UTC 2025

System load:        0.15
Usage of /:         29.5% of 23.49GB
Memory usage:       6%
Swap usage:         0%
Processes:          138
Users logged in:    1
IPv4 address for enp0s3: 10.0.2.15
IPv6 address for enp0s3: fd17:625c:f037:2:a00:27ff:fec4:9ac

* Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
  just raised the bar for easy, resilient and secure K8s cluster deployment.

  https://ubuntu.com/engage/secure-kubernetes-at-the-edge

Expanded Security Maintenance for Applications is not enabled.

Updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Sun Dec 21 01:51:56 2025 from 192.168.56.101
ramesh@rameshserver:~$ ip add show
: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:c4:09:ac brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 86256sec preferred_lft 86256sec
    inet6 fd17:625c:f037:2:a00:27ff:fec4:9ac/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86259sec preferred_lft 14259sec
    inet6 fe80::a00:27ff:fec4:9ac/64 scope link
        valid_lft forever preferred_lft forever
: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:a3:03:7a brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.101/24 metric 100 brd 192.168.56.255 scope global dynamic enp0s8
        valid_lft 456sec preferred_lft 456sec
    inet6 fe80::a00:27ff:fea3:37a/64 scope link
        valid_lft forever preferred_lft forever
```

The network setup uses VirtualBox Host-Only Networking to make a private network between the workstation and the Ubuntu Server. This keeps the server off the open internet. 192.168.56.101 is the private IP address of the Ubuntu server on the private network. This device connects to the VirtualBox host-only network, creating safe interactions between the server while the owner's machine. This setup shows how the operating system handles network connections, IP addresses, and getting data where it needs to go in a virtual environment, all while staying secure by keeping things separate

5. Using a CLI, document system specifications using ``uname``, ``free``, ``df -h``, ``ip addr``, and ``lsb_release``?

Uname -a:

```
ramesh@rameshserver:~$ uname -a
Linux rameshserver 6.8.0-90-generic #91-Ubuntu SMP PREEMPT_DYNAMIC Tue Nov 18 14:14:30 UTC 2025 x86_64 x86_64 x86_64 GNU/Linux
```

The `uname -a` search term will be used to obtain more details about the operating system of the computer. The output shows the system is using a 64-bit CPU and most fresh operating system build used by Ubuntu. Also, it shows that the kernel supports changing advance and parallel processing, both of which improve system speed and performance. The IP address that shows confirms that the commands are running on the right server.

Free -h:

```

ramesh@rameshserver:~$ sudo apt update
[sudo] password for ramesh:
Hit:1 http://gb.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://gb.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:4 http://gb.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:5 http://gb.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [175 kB]
Get:6 http://gb.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Components [212 B]
Get:7 http://gb.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [378 kB]
Get:8 http://gb.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:9 http://gb.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [7,332 B]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Get:11 http://gb.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [216 B]
Get:12 http://gb.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [10.5 kB]
Get:13 http://gb.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:14 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Get:15 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [71.4 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Fetched 1,044 kB in 2s (660 kB/s)

Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
ramesh@rameshserver:~$ free -h

```

|       | total | used  | free  | shared | buff/cache | available |
|-------|-------|-------|-------|--------|------------|-----------|
| Mem:  | 3.8Gi | 475Mi | 2.9Gi | 1.1Mi  | 638Mi      | 3.4Gi     |
| Swap: | 3.8Gi | 0B    | 3.8Gi |        |            |           |

The system's memory usage is shown in a readable format by the free -h command. just some the server's 3.8 GB of virtual memory is currently in use, relating to the output. The system is fully loaded and functioning effectively because most memory is still available. with the low demand on memory, it may be correct that no spare room is set up.

df -h:

```

ramesh@rameshserver:~$ df -h

```

| Filesystem                        | Size | Used | Avail | Use% | Mounted on     |
|-----------------------------------|------|------|-------|------|----------------|
| tmpfs                             | 392M | 1.1M | 391M  | 1%   | /run           |
| /dev/mapper/ubuntu--vg-ubuntu--lv | 24G  | 7.1G | 16G   | 32%  | /              |
| tmpfs                             | 2.0G | 0    | 2.0G  | 0%   | /dev/shm       |
| tmpfs                             | 5.0M | 0    | 5.0M  | 0%   | /run/lock      |
| /dev/sda2                         | 2.0G | 197M | 1.6G  | 11%  | /boot          |
| tmpfs                             | 392M | 12K  | 392M  | 1%   | /run/user/1000 |

Ip addr:

```
ramesh@rameshserver:~$ ip add show
: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
```

The server is set to utilise multiple network interfaces, as confirmed by the ip address display command. Internal system communication uses the return interface, and enp0s3 uses a firewall to offer internet access. An IP address that is secret is provided the enp0s8 network interface, who is utilised for safe SSH communication with the host computer. Performance and danger are made better by this online division.

lsb\_release -a:

```
ramesh@rameshserver:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04.3 LTS
Release:        24.04
Codename:       noble
```

Linux's release specifics may be seen using a the lsb\_release command. The Linux Standards Basis, or the LSB over shorter, sets uniform norms across all Linux distributions. The operating system version, release number, and password that are currently operating on the server is confirmed by this command.